

# Jesús Rodríguez González

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## PROFESSIONAL SUMMARY

Analytical and results-driven dual-degree student in Computer Engineering and Business Administration (ADE). Combining a strong foundation in advanced computation, machine learning, and statistical analysis with a strategic understanding of business operations and financial metrics. Highly proficient in Python, SQL, and C++, with hands-on experience building algorithmic solutions and data pipelines. Adept at turning complex theoretical concepts into scalable, pragmatic solutions to support data-informed product decisions.

## EDUCATION

### University of Granada

Granada, Spain

Double Bachelor's Degree in Computer Engineering and Business Administration (ADE) Exp. June 2027

- **Current GPA:** 8.5/10 (Equivalent to First-Class/Strong 2:1 Honours).
- **Specialization:** Computation and Intelligent Systems.
- **Core Coursework:** Machine Learning, Advanced Computation Models, Statistics, Corporate Finance.

## TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, C, Java, Bash.
- **Data Science & ML:** Data preprocessing, Predictive modeling, Neural Networks, Pandas, Scikit-learn.
- **Algorithms & Math:** Metaheuristics (AGG, AGE, AM), Complexity Theory, Statistics, Probability.
- **Infrastructure & Tools:** Linux (Arch, Ubuntu), Docker, Prometheus, Grafana, Git.
- **Languages:** English (C1 Advanced Cambridge Certification), Spanish (Native).

## RELEVANT PROJECTS

### Olist Growth Engine (Churn Prediction & Dynamic Pricing)

*Python, PostgreSQL, Streamlit*

- Developed an end-to-end predictive SaaS model to forecast customer churn and generate dynamic pricing strategies.
- Engineered the data architecture using PostgreSQL and Python, implementing machine learning models focused on retention value.
- Built an interactive Streamlit dashboard designed to translate complex algorithmic probabilities into clear, actionable commercial decisions for stakeholders.

### Machine Learning & Predictive Modeling Practices

*Python, Scikit-learn, Pandas*

- Developed and trained multiple machine learning models across various datasets to solve classification and regression problems.
- Implemented complete data pipelines including data cleaning, feature engineering, and exploratory data analysis (EDA) to prepare sets for training.

### Financial Portfolio Optimization via Metaheuristics

*C++, Algorithmic Design*

- Engineered advanced evolutionary and memetic algorithms (AGG, AGE, AM) from scratch to solve highly constrained portfolio optimization problems.
- Simulated investment scenarios to maximize yield while balancing computational complexity, bridging theoretical computer science with practical financial applications.

### Advanced Computation & Complexity Theory

*Formal Logic, NP-Completeness*

- Authored a formal mathematical proof demonstrating the NP-Completeness of a complex computational problem.
- Applied L-space and strict complexity theory principles to evaluate algorithmic limitations, showcasing strong problem-solving capabilities and analytical rigor.

## **Systems Monitoring & Observability**

*Prometheus, Grafana, Linux*

- Configured observability pipelines utilizing Prometheus and Grafana to track and visualize infrastructure metrics.
- Developed automated Bash scripts in advanced Linux environments to optimize workflow efficiency and maintain data infrastructure reliability.